

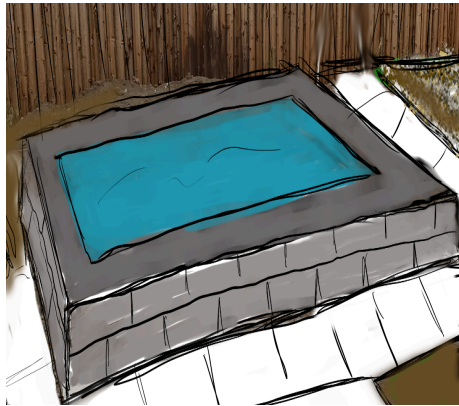
'12 x '16 x '4 CMU WALL SWIMMING POOL

Proposed construction under a Homeowner-Affidavit permit; of a swimming pool utilizing Concrete Masonry Unit (CMU) wall construction set upon a reinforced concrete slab. The CMUs, will be partially earth-retained, are to receive a multi-layered finish system with below-grade waterproofing, cementitious plaster, and a durable, pool-compatible coating. A filtration and sanitation system will be integrated, and designed to current industry standards. Hydrostatic pressure relief and groundwater management measures are included. A separate 5' x 3' concrete equipment pad, is proposed approximately '12 from the primary pool structure to house essential plumbing and mechanical components. All subterranean plumbing shall be placed within properly excavated trenches, under frost line, backfilled with free-draining material and a sand layer, strictly adhering to all relevant codes. All reinforcing steel will maintain 6" clearance from earth and formwork prior to pour, and will have a earth bonding system. Two faces of the main structure are to have a concrete walkway that is to be embedded into the CMUs with epoxy. Project is to use existing electrical that is located in proximity to proposed equipment pad and $\geq$ '12 from proposed swimming pool.

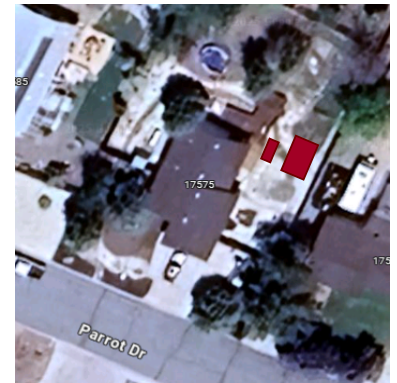
Contents

G - General / Cover - 1.01
V - Site Plan / Layout - 1.01
S - Structural / Overview - 3.01
S - Structural / Main Slab - 3.02
S - Structural / Block Layout - 3.04
S - Structural / Pump Pad - 3.11
S - Structural / Walkway- 3.12
B/C - Geotechnical / Civil - 1.01
G - General / Elevation - 2.01
P - Plumbing / Overview - 1.01

Artistic



Example



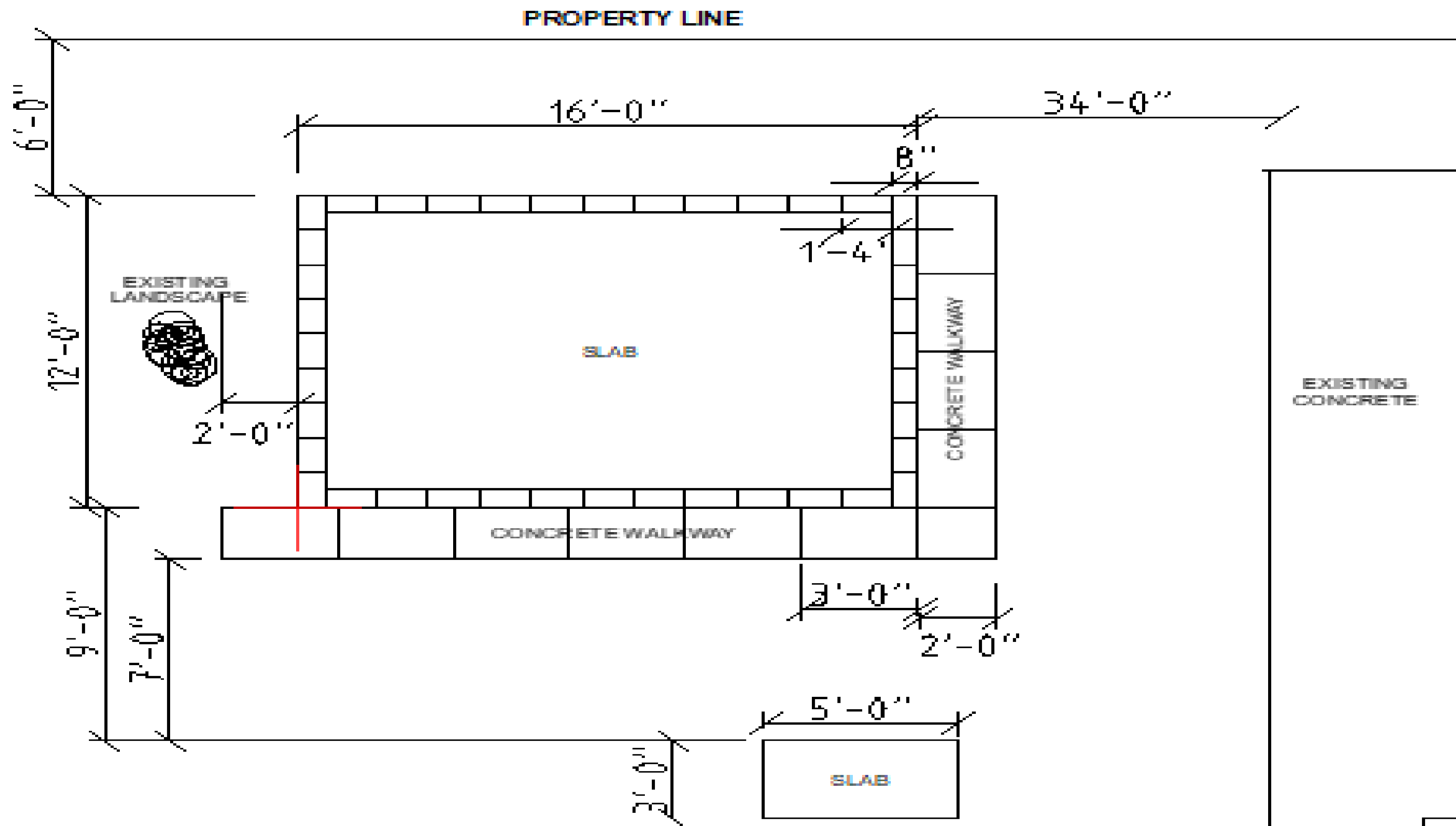
Approximate proposed location

CMU POOL

Cruz Wooten
17575 Parrot Drive
Reno, Nevada
775.501.4398

Cover
G1.01

Drawing
Page
/



SITE PLAN

- * '12 x '16 x 6" MAIN SLAB (BELOW GRADE)
- * '5 x '3 x 6" SECONDARY SLAB (AT GRADE)
- * CONCRETE WALKING PATH
ALONG SOUTH/WEST FACES
WITH '3 x '2 x 4" FLAGS
- * '6 PROPERTY LINE SETBACK
- * CMU WALLS ON MAIN SLAB
- * CMU DIMENSIONS: 8" x 8" x 16"
- * 4" PERFORATED PVC DRAIN UNDER MAIN SLAB
- * #4 REBAR FOR BOTH SLABS
- * MAIN SLAB HAS COMPACTED FILL AND VAPOR/MOISTURE
BARRIER INSTALLED ON TOP OF DRAINAGE TRENCH/PIPE
- * NOT TO SCALE *

CMU POOL

Cruz Wootten
17575 Parrot Drive
Reno, Nevada
775.501.4398

Survey
V1.01

Drawing
Page
/

Structural Overview

Structural 3.01

1 – Water volume

Volume (cu ft) = length × width × depth = 16 × 12 × 4.

Total gallons = 5,745.03936

2 – Concrete volume

Footing

Perimeter length = 56 ft.

Footing cross-section = 16" × 8" = (16/12) ft × (8/12) ft.

Footing total volume = 1.844 yd³

Main Slab

Area = 12 × 16 = 192 sq ft.

Thickness = 6" = 0.5 ft.

Volume = area × thickness = 192 × 0.5 = 96 cu ft.

Slab total volume = 3.556 yd³.

3 – Slab Grid

Slab reinforcement: #4 bars @ 8" o.c. both directions (placed mid-depth).

Number of bars/ft long-direction (parallel to long side) = 208 ft.

Number of bars/ft short-direction = 204 ft.

Total linear feet of slab #4 = 208 + 204 = 412 ft.

4 – Summary

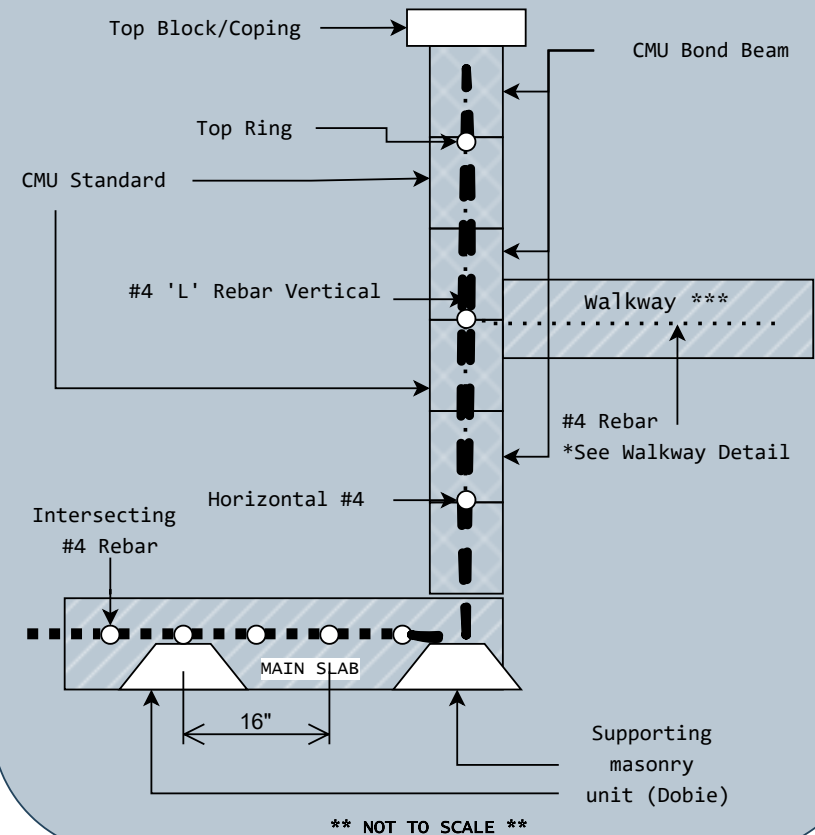
Footing: 2 #4 continuous (perimeter) – total ≈ 112 ft.

Bond beam: 1 #4 horizontal continuous – 56 ft (specify if 2 required).

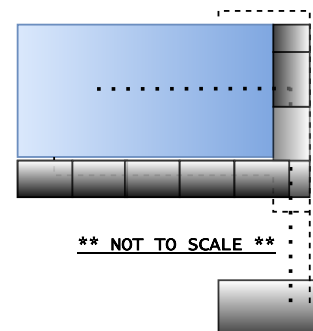
Vertical wall: #4 @ 24" o.c. – 28 bars, total ≈ 149.33 ft.

Slab: #4 @ 12" o.c. both directions – total ≈ 412 ft.

Cross Section Overview



Overview



CMU POOL

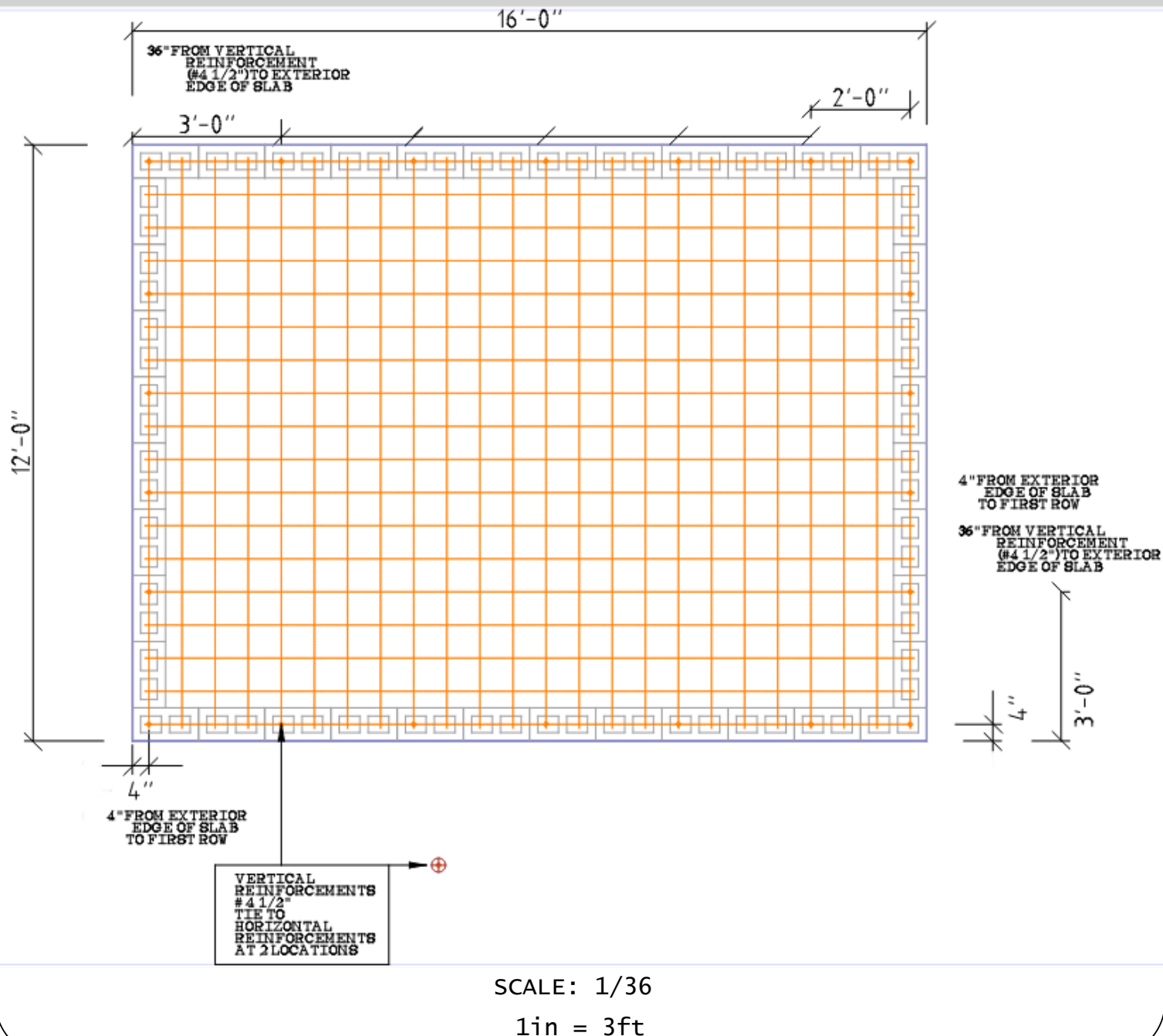
Cruz Wootten
17575 Parrot Drive
Reno, Nevada
775.501.4398

Structural
S3.01

Drawing
Page
/

Structural 3.02

MAIN SLAB REINFORCEMENT



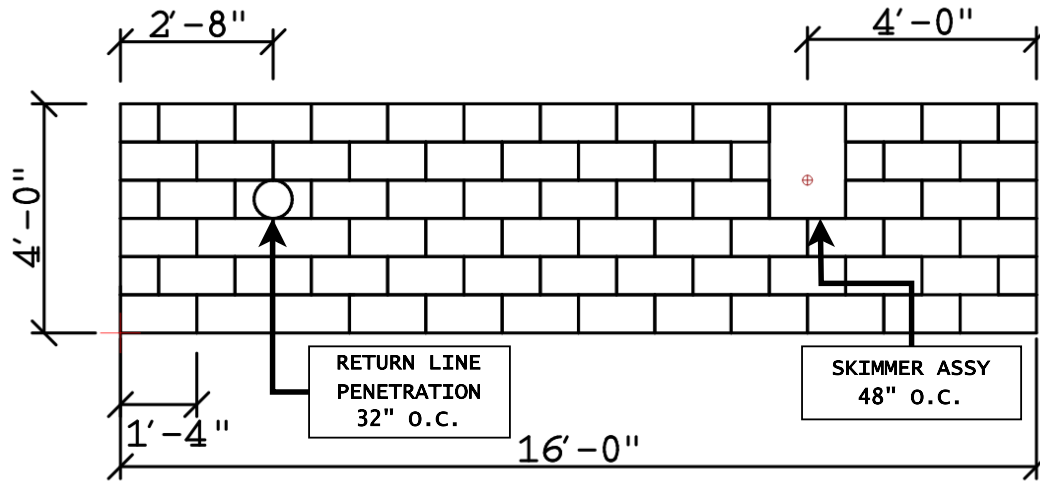
CMU POOL

Cruz Wootten
17575 Parrot Drive
Reno, Nevada
775.501.4398

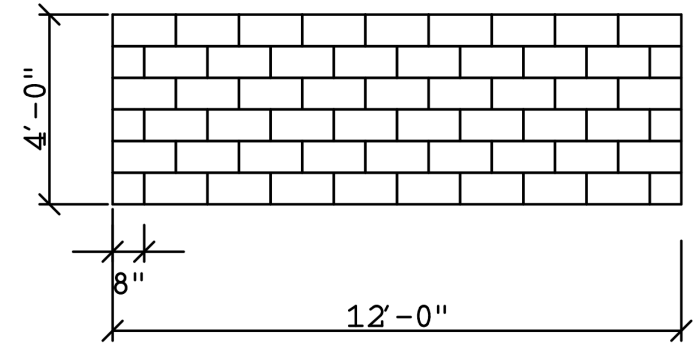
Structural
S3.02

Drawing
Page
/

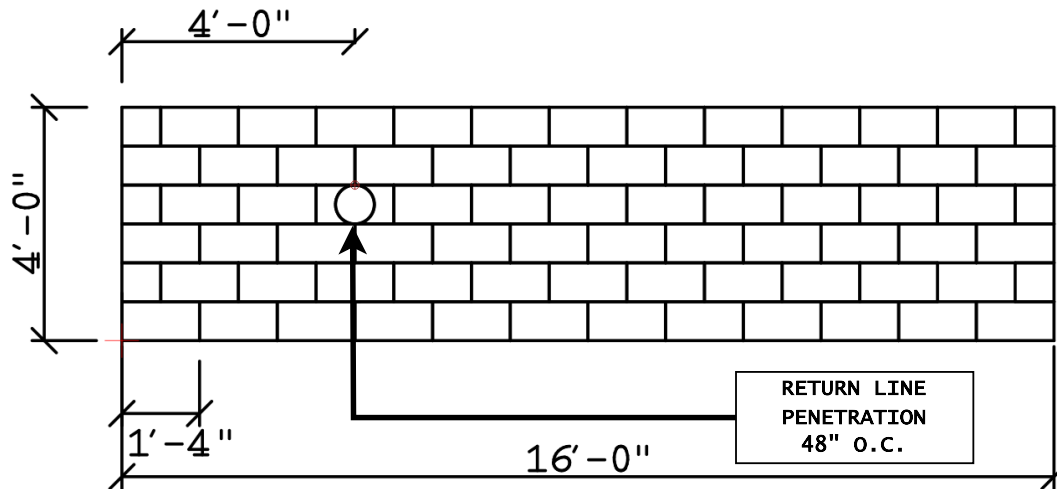
Block Layout (East Elevation)



Block Layout (North & South Elevations)



Block Layout (West Elevation)



Structural 3.04

- * Standard CMU for first course, third course, and fifth course
- * Bond beam CMU for second course, fourth course, and sixth course
- * Skimmer assembly to be 48" from right edge on East Elevation
- * Return line on East Elevation is 32" O.C. from left edge
- * Return line on West Elevation is 48" O.C. from left edge
- * Vibrated concrete for CMU cells on all vertical walls

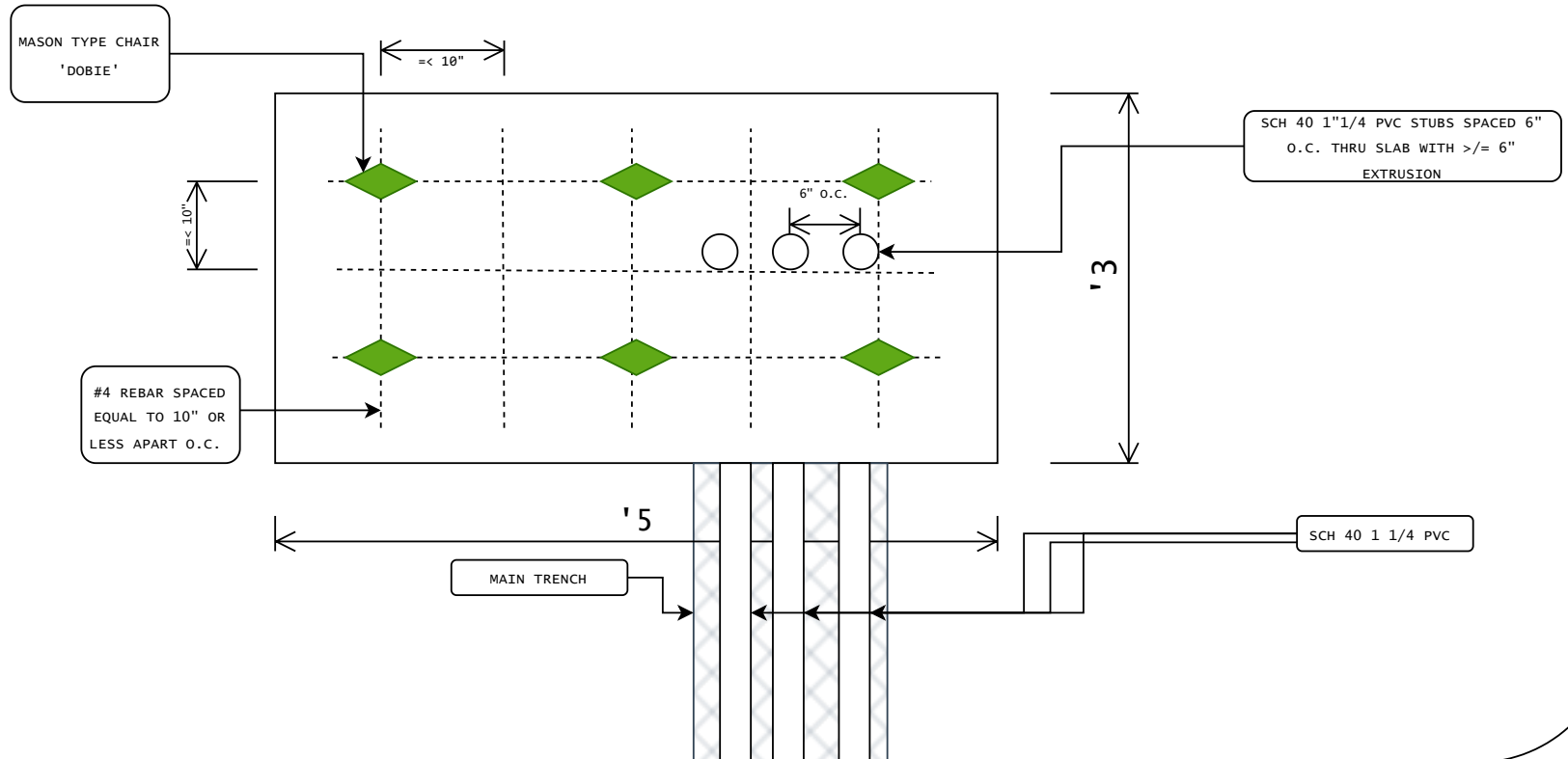
CMU POOL

Cruz Wooten
17575 Parrot Drive
Reno, Nevada
775.501.4398

Structural
S3.04

Drawing
Page
/

SECONDARY SLAB (Pump Pad)



Structural 3.11

- * All slabs to use #4 Rebar
- * Seperate plumbing stubs 6" O.C.

**** NOT TO SCALE ****

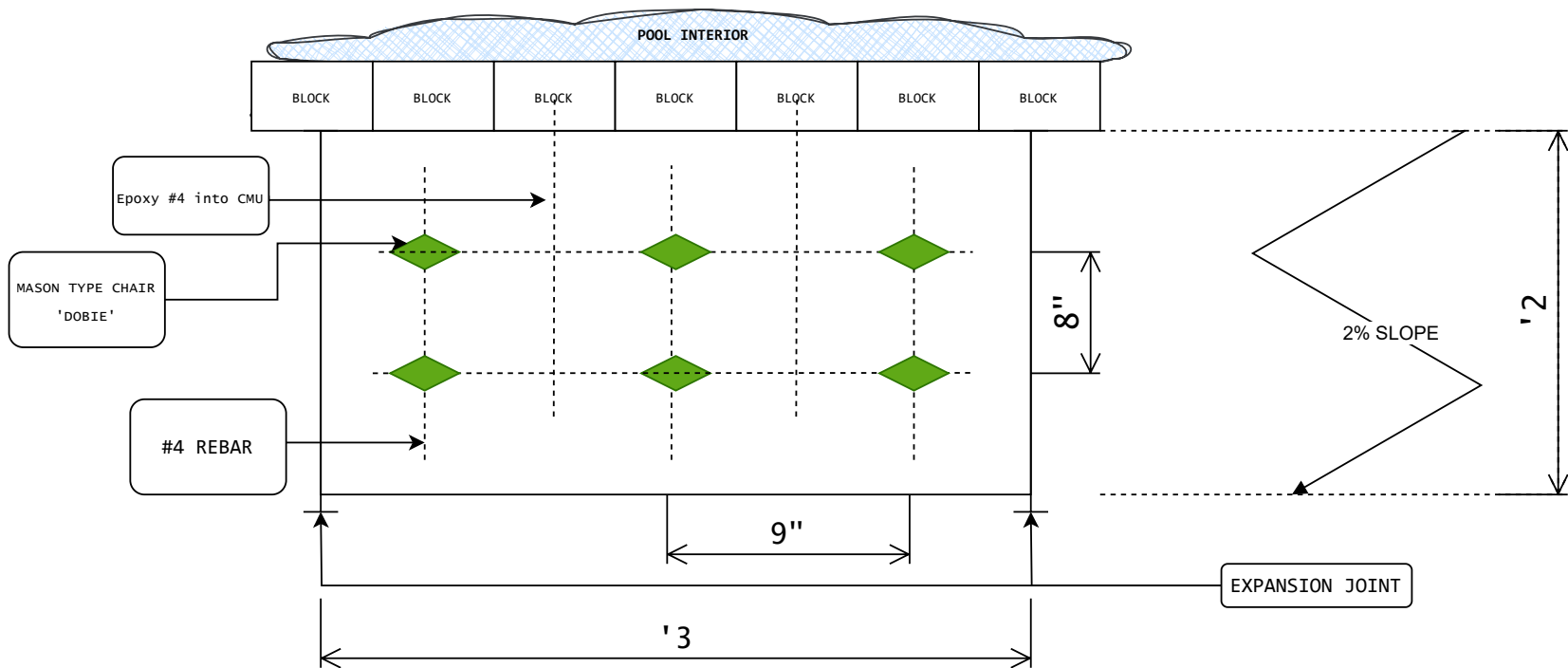
CMU POOL

Cruz Wootten
17575 Parrot Drive
Reno, Nevada
775.501.4398

Structural
S3.11

Drawing
Page
/

WALKWAY DETAIL (SINGLE FLAG)



Structural 3.12

- * All slabs to use #4 Rebar
- * Repeat this detail for every flag

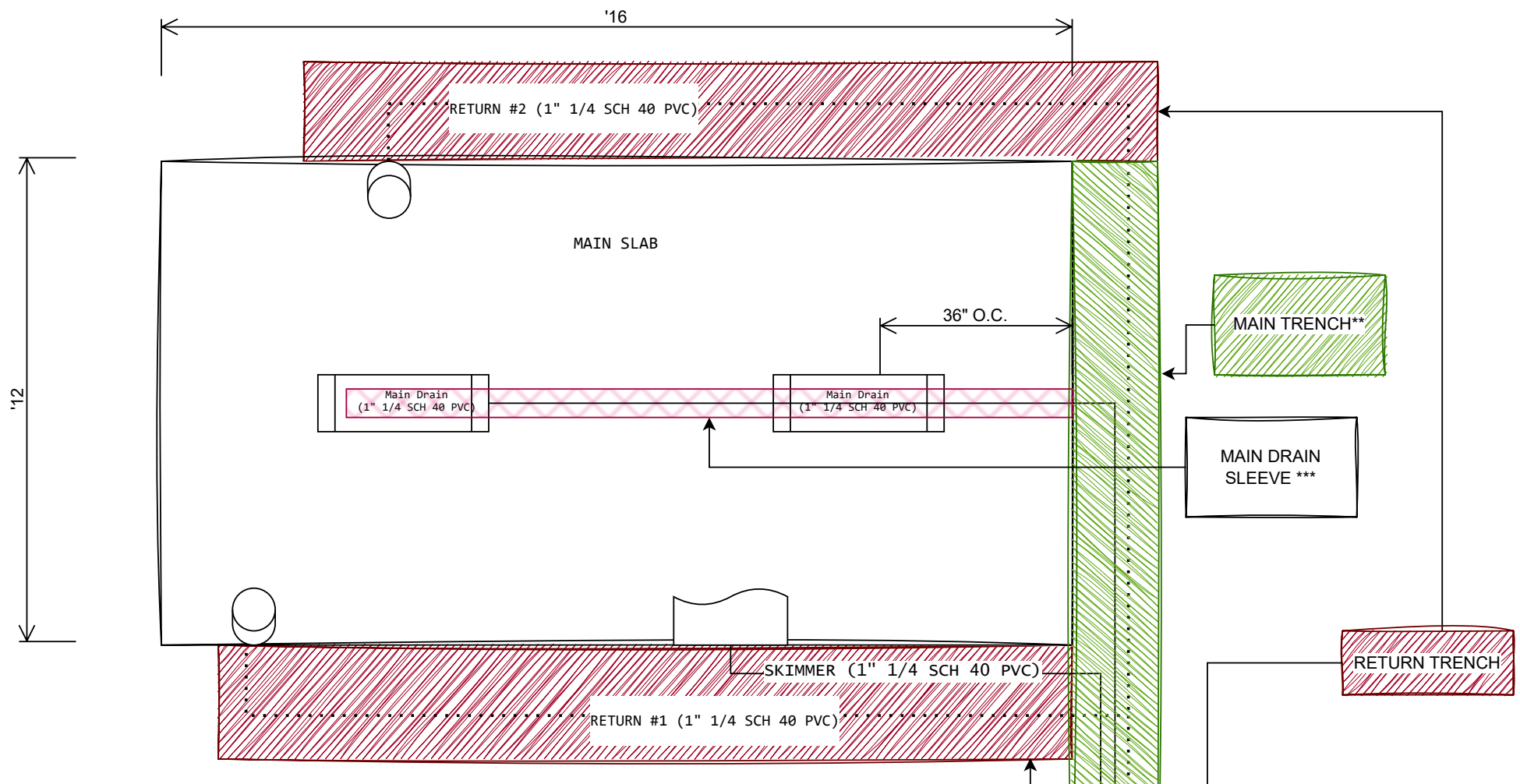
**** NOT TO SCALE ****

CMU POOL

Cruz Wootten
17575 Parrot Drive
Reno, Nevada
775.501.4398

Structural
S3.12

Drawing
Page
/



Plumbing Overview 1

————— Solid Line denotes Suction Side of the system

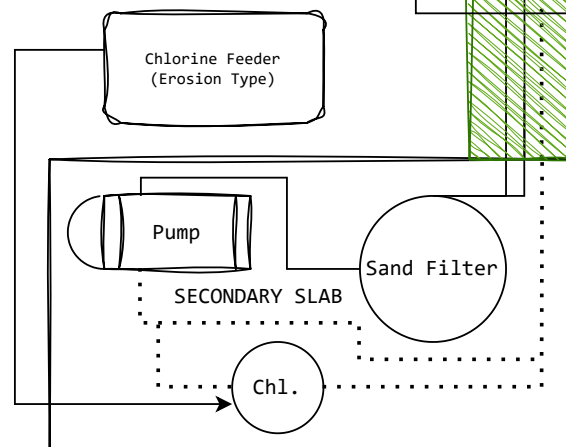
----- Dotted Line denotes Pressure Side of the system

* System to be piped in SCH 40 PVC 1" 1/4 in compliance with local code requirements.

** Trenches are to be below frostline, with a base of permeabl fill then topped with sand.

*** Portion of Main Drain line under slab to be sleeved inside of 4" Perforated PVC for hydrostatic relief and water table changes

**** NOT TO SCALE ****

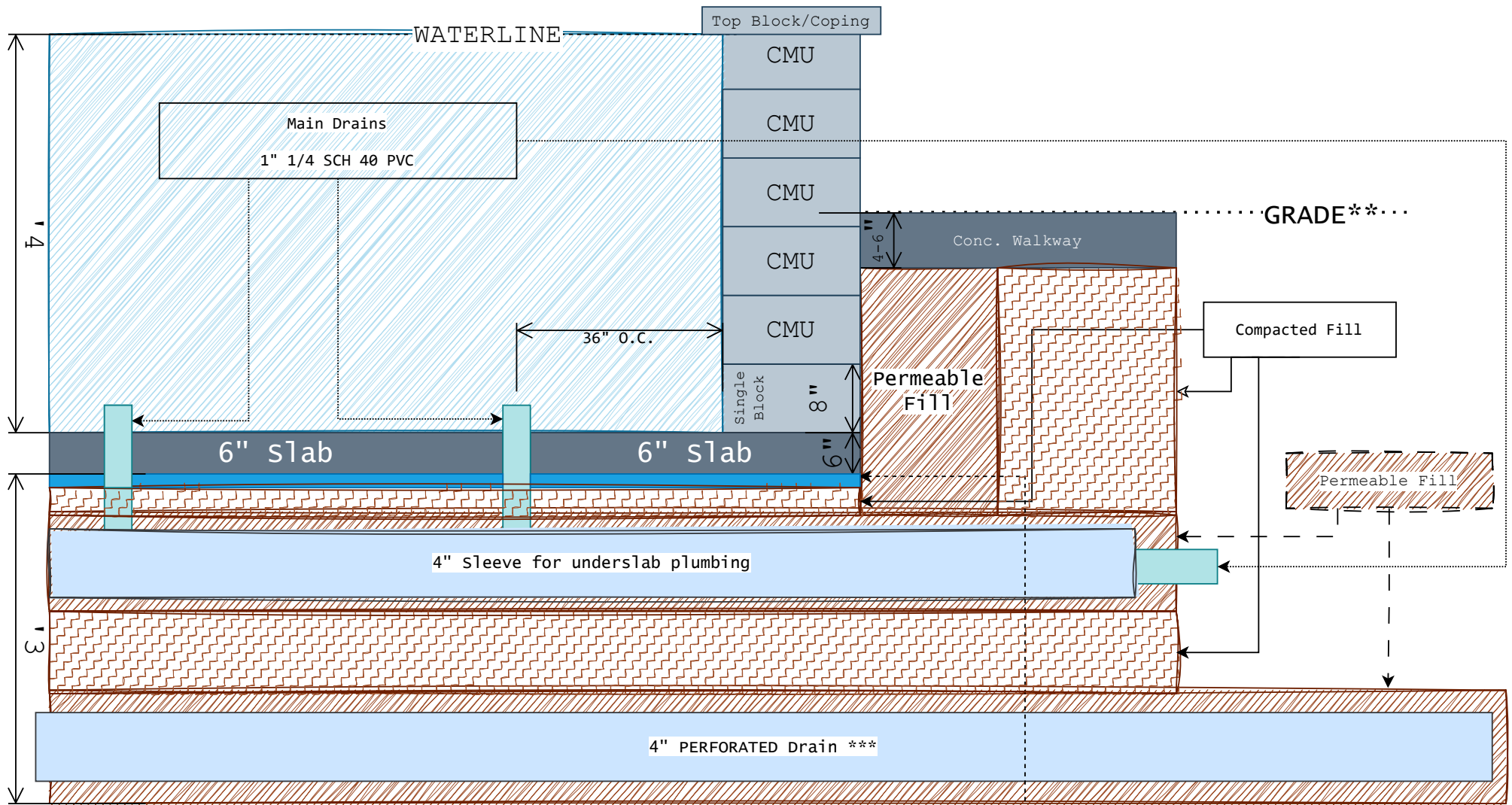


CMU POOL

Cruz Wootten
17575 Parrot Drive
Reno, Nevada
775.501.4398

PLUMB
P1.01

Drawing
Page
/



Overview: Cross Section (East Elevation)

* Main Drain piping to run inside 4" sleeve under slab to trench

** Grade will vary

*** 4" drain to run under slab for hydrostatic relief, saturated earth and site drainage (See Geo/Civil)

**** Minimum 10 mil sheeting

(Not To Scale)

Vapor/Moisture
Underlayment ****

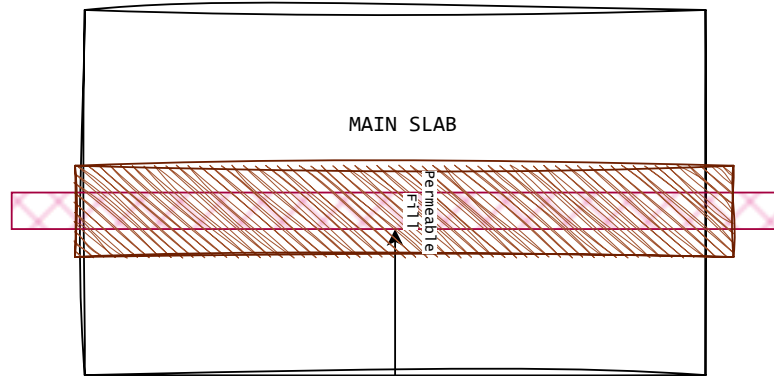
CMU POOL

Cruz Wootten
17575 Parrot Drive
Reno, Nevada
775.501.4398

General
G2.01

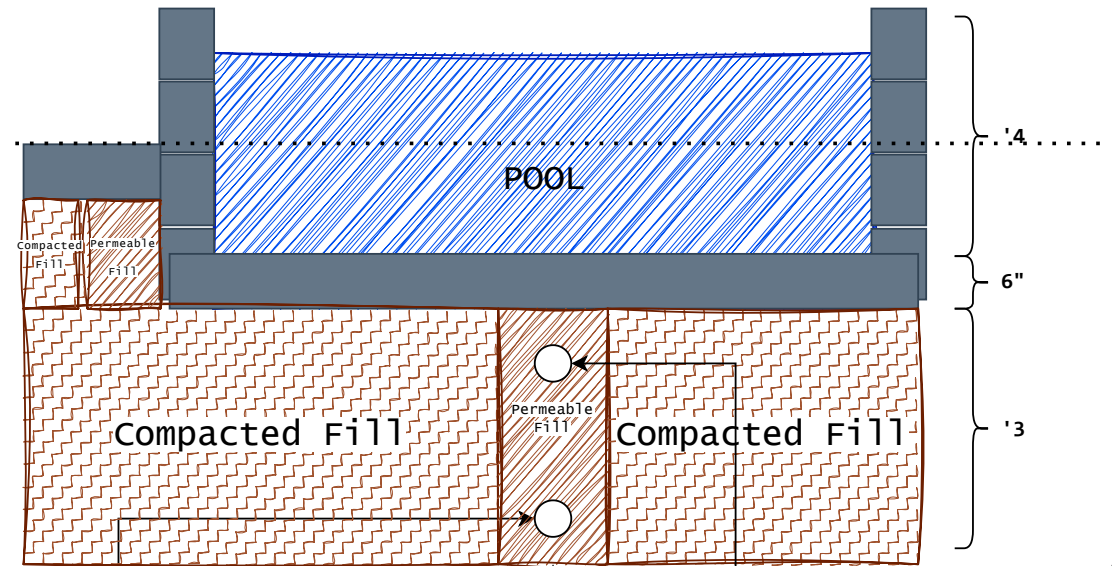
Drawing
Page
/

OVERVIEW



Perforated 4" PVC Drain
* "FRENCH DRAIN" for ground water

CROSS SECTION



Main Drain Trench

Main Drain for
filtration system

Geotechnical / Civil (Drainage)

- * 4" Perforated PVC Drain to run under entire main slab for groundwater control
- * Surround Drain with Permeable Fill for hydrostatic relief

**** NOT TO SCALE ****

CMU POOL

Cruz Wootten
17575 Parrot Drive
Reno, Nevada
775.501.4398

Geo/Cvl
B/C 1.01

Drawing
Page
/